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ACME PETROCHEMICAL DEMO COMPLEX - UNIT 2300

PROCESS DATA SHEET: REACTOR / DECOMPOSER DRUM

Document Code: DS-D2304 | Revision: Z1 | Status: Issued for Design

Equipment Tag: D-2304

Equipment Name: Decomposer Reactor (Acid-Catalyzed CHP Cleavage Drum)

Equipment Class: Chemical Reactor / Pressure Vessel

Process Unit: Unit 2300 (Decomposition & Recovery Section)

Service: Exothermic Acid-Catalyzed Cleavage of Cumene Hydroperoxide (CHP) to Phenol and Acetone

1. AUTHORITATIVE MECHANICAL & DESIGN PARAMETERS (DS-D2304 Rev Z1)

- Internal Design Pressure: 11.0 kg/cm²g
- Internal Design Temperature: 250 degC
- Normal Operating Pressure: 1.8 kg/cm²g
- Normal Operating Temperature: 82 degC
- Vessel Inside Diameter (ID): 2400 mm
- Vessel Tangent-to-Tangent Length (T/T): 6800 mm
- Shell Material of Construction: SA-240 Type 316L Stainless Steel
- Corrosion Allowance: 1.5 mm
- Mechanical Volume: 31.8 m³
- Heat of Reaction (CHP Cleavage): -1580 kJ/kg CHP (Highly Exothermic)

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DS-D2304 PAGE 2 OF 4 - PROCESS CONNECTIONS & RECIRCULATION LOOP

Equipment Tag: D-2304 (Decomposer Reactor)

- Feed Inlet (Stream S229): Concentrated 82 wt% CHP from Preflash Column V-2301 via Pump P-2301A/B (14,500 kg/hr)
- Catalyst Feed (Stream S310): Dilute Sulfuric Acid Catalyst (180 ppmw H₂SO₄ in Acetone) via Valve XV-1305
- Recirculation Outlet (Stream S379): 185,000 kg/hr reaction mixture to Circulation Pump P-2304A/B and Reactor Co
- Cooled Recirculation Return (Stream S380): Returns from Reactor Cooler E-2307 at 74 degC to maintain 12:1 recyc
- Reactor Product Effluent (Stream S384): 14,500 kg/hr cleaved Phenol/Acetone product to Neutralization Drum D-23
- Emergency Quench Inlet (Stream S387): Cold Cumene Quench injection via Emergency Valve XV-1309

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DS-D2304 PAGE 3 OF 4 - SAFETY INSTRUMENTATION & OVERPRESSURE PROTECTION

Equipment Tag: D-2304

- Guided-Wave Radar Level Transmitter LT-1301: Range 0 to 4500 mm (Normal Level 55%, LSHH 82%, LSLI 20%), modulated
- SIL-2 Voting Temperature Switch High-High TXSHH-1301 (2oo3 voting TXSHH-1301A/B/C):
 - * High Alarm (TAH): 92.0 degC
 - * High-High Trip Setpoint (TXSHH): 115.0 degC -> Activates Interlock I-2304 (closes CHP feed, trips XV-1305 ac
- Pressure Transmitter PT-1301: Range 0 to 15.0 kg/cm2g (PAH at 3.2 kg/cm2g)
- Pressure Safety Valves PSV-2304A / PSV-2304B:
 - * Set Pressure: 11.0 kg/cm2g (matches DS-D2304 Internal Design Pressure of 11.0 kg/cm2g)
 - * Orifice Size: 4J6 (API 526)
 - * Relieving Case: Blocked Outlet / Runaway Two-Phase CHP Decomposition to Knock-Out Drum D-2320

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DS-D2304 PAGE 4 OF 4 - NOZZLE SCHEDULE & NOTES

Equipment Tag: D-2304

Nozzle	Size	Rating	Service	Connected Equipment / Line
N1	4 in	300# RF	Concentrated CHP Feed	Stream S229 from V-2301 / P-2301A/B
N2	10 in	300# RF	Circulation Draw to Cooler	Stream S379 to P-2304A/B & E-2307
N3	10 in	300# RF	Cooled Recycle Return	Stream S380 from E-2307
N4	4 in	300# RF	Cleavage Product Outlet	Stream S384 via LV-1301 to D-2310
N5	4 in	300# RF	Emergency Quench Inlet	Stream S387 via XV-1309
N6	6 in	300# RF	Relief Header Nozzle	PSV-2304A/B to D-2320